

Data for project

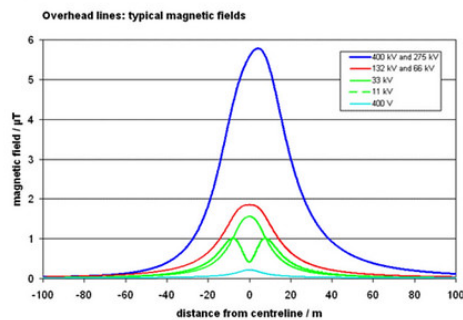
- Weather : sunny, mid-day (no shadow), no wind
- Safety Distance for UAV Inspection around Extra-High Voltage Transmission Lines

Table 1

Range of sideline protection zones for transmission at all levels in general areas.

Voltage level (KV)	Outside distance (L_s) (m)
66–110	10
220–330	15
500	20
750	25

Overhead power lines



Electricity is carried round the country on overhead lines at various voltages. All overhead lines produce electric and magnetic fields. The field is highest directly under the line and falls to the sides.

This graph shows typical magnetic fields. [See also similar graphs](#) for electric fields and for maximum fields

Typical ground-level UK field levels from overhead power lines

		Magnetic Field (microteslas)	Electric Field (volts per metre)
The largest steel pylons (275 kV and 400 kV)	Maximum field (under line)	100	11,000
	Typical field (under line)	5-10	3000-5000
	Typical field (25 m to side)	1-2	200-500
Smaller steel pylons (132 kV)	Maximum field (under line)	40	4,000
	Typical field (under line)	0.5 – 2	1000-2000
	Typical field (25 m to side)	0.05-0.2	100-200
Wooden poles (11 kV and 33 kV)	Maximum field (under line)	7	700
	Typical field (under line)	0.2-0.5	200
	Typical field (25 m to side)	0.01-0.05	10-20

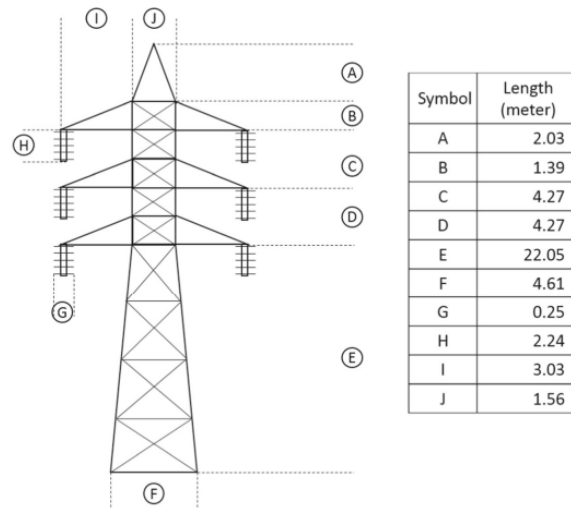
Safety distance between drones and power tower : 10m

<https://www.hindawi.com/journals/mpe/2021/8894055/>

<https://www.mdpi.com/2504-446X/8/2/47>

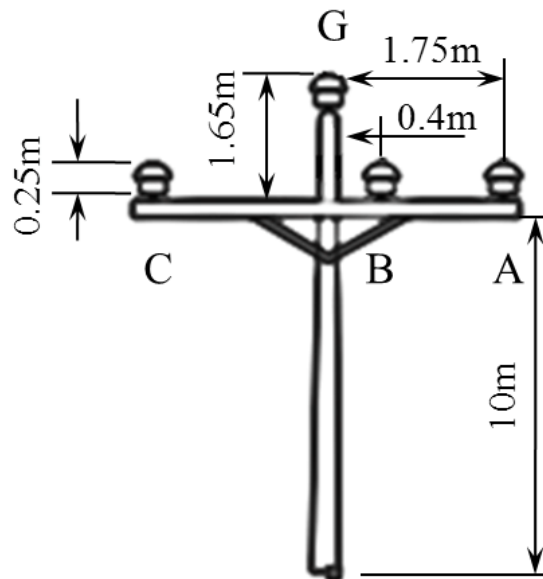
- Tower dimension

<https://link.springer.com/article/10.1007/s10846-018-0947-9>



Dimension for 132 kV transmission tower

Transform this dimension into rectangular dimension : 34,01m high x 7.62m wide



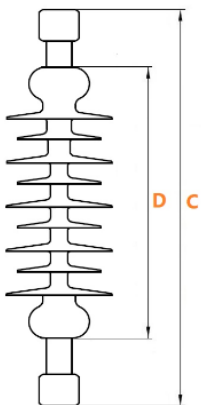
Dimension for 22 kV transmission tower

Dimension of the tower : 11.65m high x 3.75m wide

- Area that we need to go

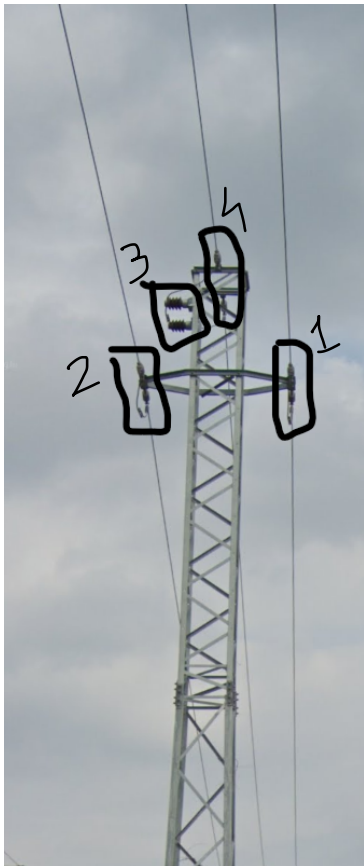
Each insulator and earthing rope

- insulator



Total length C	1540 mm
Insulating length D	1340 mm

<https://www.hsp-online.de/Product/composite-insulator-long-rod-insulator-132-kv-120-kn-5200-mm-1540-mm/>



- 1 Left insulator : 10m high x 1.75m wide
- 2 Right insulator : 10m high x 1.75m wide
- 3 Middle right : 10.8m high x 0.8m wide
- 4 Middle left : 11.65m high x 0 wide

- **Camera**

<https://laprotectionvideo.fr/fr/drones/881-drone-industriel-avec-camera-3760284830720.html>

Caméra

Capteur	1 / 1.9 "CMOS STARVIS™
Pixel efficace	2560 (H) x1920 (V), 6 mégapixels
Vitesse d'obturation électronique	1 / 1s ~ 1 / 30,000s
Distance focale	6mm ~ 180mm
Angle de vue	H: 61,2° ~ 2,1°
Zoom optique	30x

=> If the focal length = 15mm and you're 10m from the object, the object can have a maximum length of 3m to be in the picture. So at 10m, the insulator will represent 1/3 of the photo.

<https://support.pix4d.com/hc/en-us/articles/202559809-Ground-sampling-distance-GSD-in-photogrammetry>

- **UAV**

<https://link.springer.com/article/10.1007/s10846-018-0947-9>

UAS specifications	
Flying speed (FS)	5 m/s
Hovering time to take an image (H)	3 s
Maximum flight time (FT^{max})	491 s
Maximum flight distance (FD^{max})	1,856 m

Hovering time to take an image : 3s

- **Distance between drones : 5m**

Document can be used for introduction of the presentation

<http://mydocs.epri.com/docs/corporatedocuments/sectorpages/pdu/sensorsrobots/Tline%20Roadmap.pdf>